Goldman Sachs Research

Equity Research

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Americas Technology: Hardware: Data center & campus networking preview

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We update our industry networking model to reflect the latest C1Q26 data from 650 Group. 650 Group upgraded its outlook for the AI Ethernet data center switching market, where spending should reach ~$89 bn in 2030, growing at a 61% 2025-30 CAGR with stronger demand across all customer verticals (hyperscalers, tier 2 cloud, and enterprise). We view the outlook for increased networking spend as a result of the growing complexity of AI data center builds which should require greater networking attach to support high-performance, low-latency data traffic flows. Within the campus market: (1) WLAN (indoor + cloud)'s 2026-2030 outlook was raised by ~10% on average (now expecting ~$13.4 bn in 2030, growing at a ~7% 5-yr CAGR v. ~$10.6 bn prior), primarily driven by higher WiFi 7 estimates (+13% on average across 2026-2030); (2) Enterprise switching estimates grew by ~3% on average to reach ~$28 bn by 2030 (v. $27 bn prior). We view upgraded campus spending outlooks as a result of (a) continued need for network modernization to support greater network data traffic flows, as well as (b) partial impact of memory price increases (for campus switches). **We highlight market share shifts by equipment market below.**

Set up into C2Q26 earnings: Across our networking coverage, we believe ANET & CLS are well set up into earnings. Quarter-to-date, CLS (+26% in C2Q26) and ANET (+36% in C2Q26) underperformed the rest of our networking coverage (CSCO +51%, HPE +88%, FFIV +41%). We believe the relative underperformance of these two companies may be in part due to supply chain constraints on switching silicon that limits near-term revenue upside to guidance. However, we believe demand for AI Ethernet data center networking switches, for which spending by hyperscalers (their largest customers) should grow by ~10X by 2030 (Exhibit 1), should be non-perishable and further benefit 2028/29 growth. For HPE, we remain constructive on the positive synergies in HPE's campus portfolio with stable market share in WLAN and campus switching in the 3 quarters post acquisition of Juniper, and continue to see upside to F2027 networking outlook (+8-12% revenue growth) on strong industry trends in data center switching & DCI.

Data center networking forecast updates

Exhibit 1: Global data center switching spend is expected to grow robustly, quadrupling by 2030 (relative to 2025)

Global data center switching spend by vertical and AI/traditional ($, mn)

($, mn)	AI Data Center Switching*		
	2025	2030	5-yr CAGR
Hyperscalers	$4,948	$52,717	61%
Tier 2 Cloud + SP	$2,606	$28,249	61%
Enterprise	$553	$7,933	70%
Total AI Data Center	**$8,107**	**$88,899**	**61%**

	Traditional Data Center Switching		
	2025	2030	5-yr CAGR
Hyperscalers	$7,708	$7,573	0%
Tier 2 Cloud + SP	$7,013	$18,497	21%
Enterprise	$8,960	$9,570	1%
Total Traditional Data Center	**$23,682**	**$35,640**	**9%**

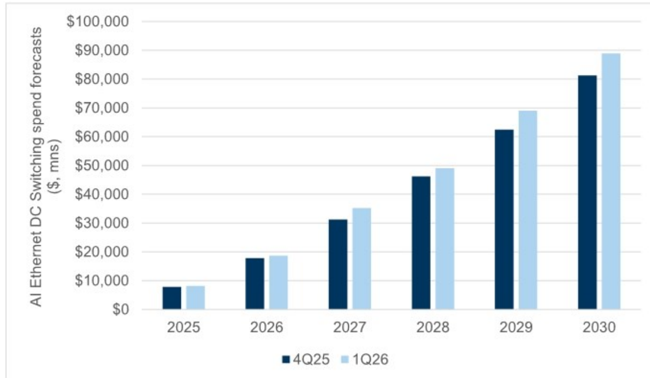
	Total Data Center Switching*		
	2025	2030	5-yr CAGR
Hyperscalers	$12,656	$60,289	37%
Tier 2 Cloud + SP	$9,619	$46,746	37%
Enterprise	$9,514	$17,503	13%
Total Data Center	**$31,788**	**$124,538**	**31%**

Includes Ethernet (frontend, backend) switches

Source: 650 Group

Exhibit 2: 650 Group's AI data center switching 2026-2030 outlook was raised by 9% on average annually

Industry AI data center switching estimate revisions, ($, mn)

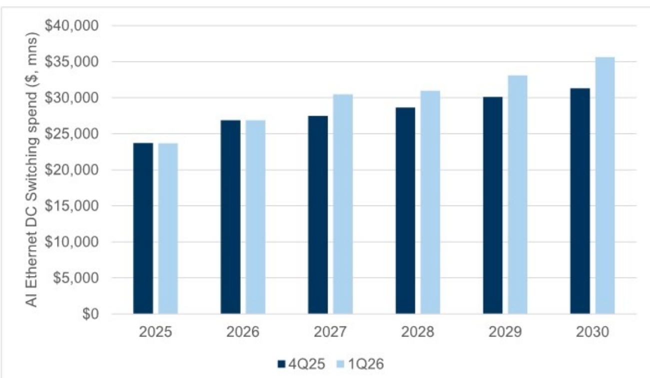


Only includes AI Ethernet Front-end & Back-end switching

Source: 650 Group

Exhibit 3: The outlook for the rest of the data center switching market (excluding AI Ethernet) was revised upwards by 9% on average as well for 2026-2030

Non-AI Ethernet DC switching estimate revisions ($, mn)

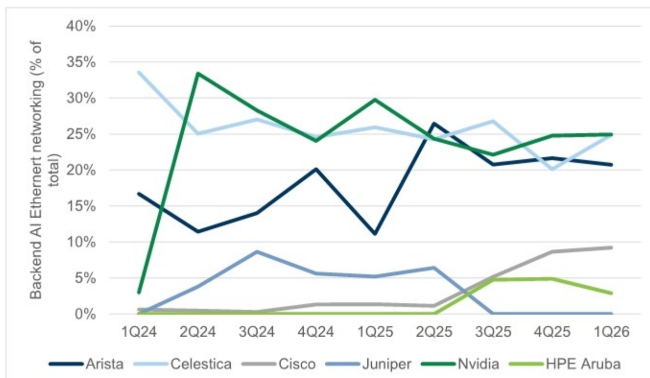


Excludes AI Ethernet front end & back end switching

Source: 650 Group

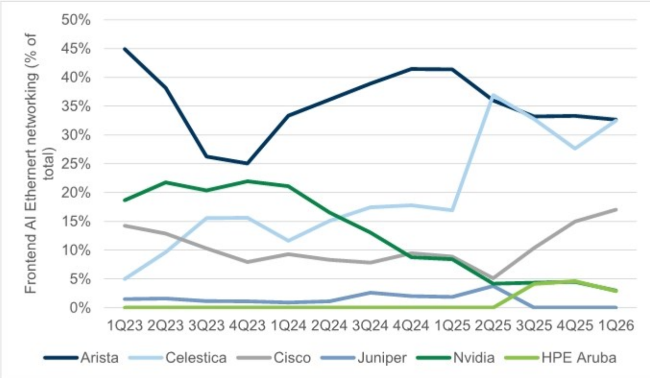
Exhibit 4: Celestica and Nvidia lead the back-end AI Ethernet networking market, with Arista closely following

Backend AI Ethernet networking revenue market share (%)



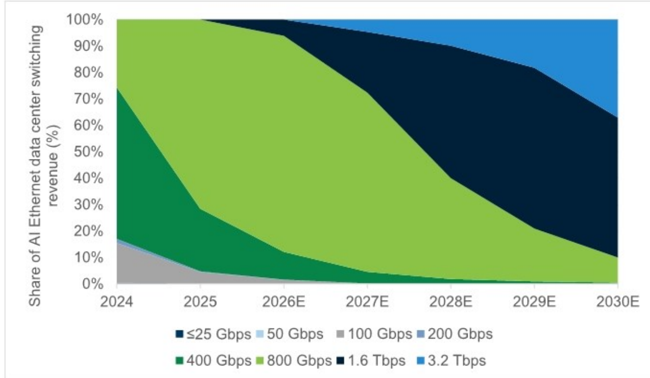
Source: 650 Group

Exhibit 5: Celestica and Arista lead the front-end networking market with 30%+ share, though Cisco has gained 12pp of share over the past 3 quarters
Front-end AI Ethernet networking revenue market share (% of total)



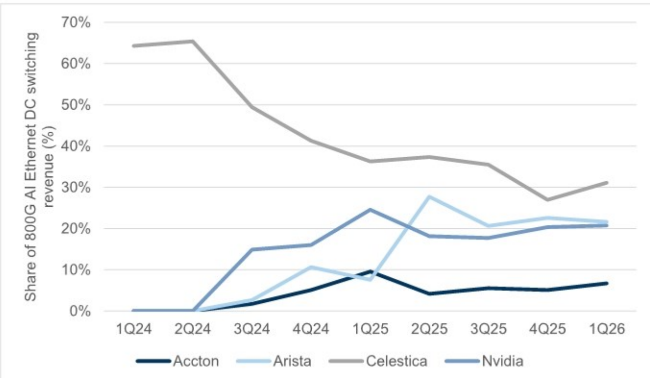
Source: 650 Group

Exhibit 6: 800G should comprise ~80% of total AI DC Ethernet Switching revenue in 2026, before 1.6T begins to meaningfully ramp in 2027
Share of AI Ethernet data center switching revenue (%) by port speed



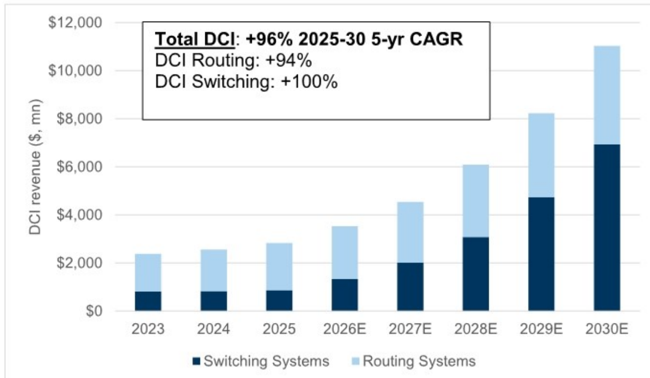
Source: 650 Group

Exhibit 7: Celestica continues to lead the 800G switching market, though Arista and Nvidia have gained share over the past year
Share of 800G switching revenue (%)



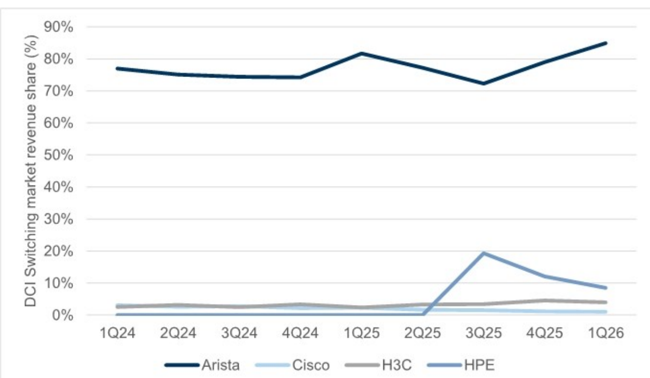
Source: 650 Group

Exhibit 8: DCI spending growth, nearly a 100% 5-yr CAGR through 2030, should be roughly evenly distributed across switching and routing
DCI revenue by switching vs routing



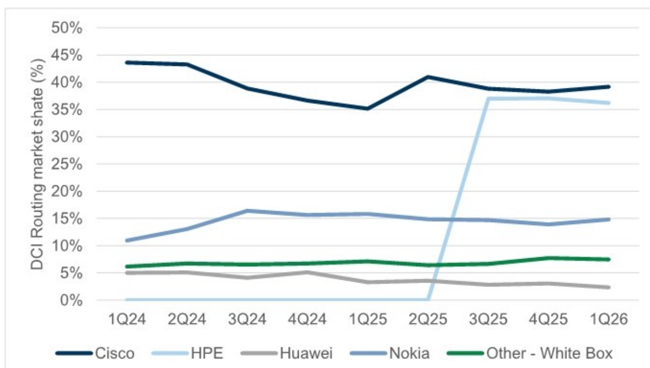
Source: 650 Group

Exhibit 9: Arista strongly leads the overall DCI switching market
DCI switching market revenue share by vendor (%)



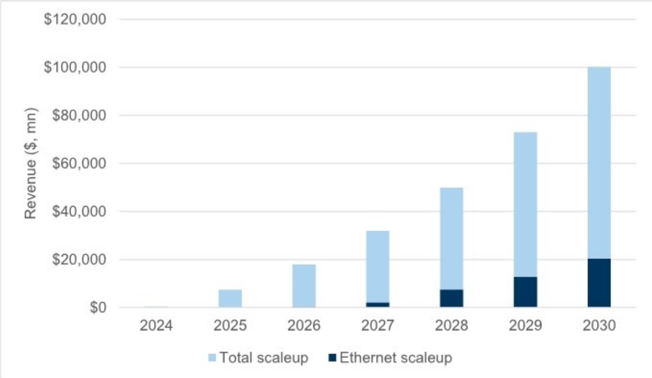
Source: 650 Group

Exhibit 10: Cisco & HPE closely lead the DCI routing market with 39% and 36% of market revenue respectively
DCI revenue market share (%)



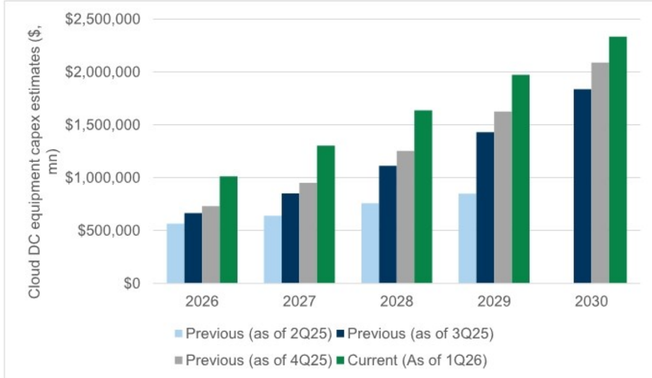
Source: 650 Group

Exhibit 11: The overall scale-up switching market should grow from ~$7 bn in 2025 to ~$100 bn in 2030, with Ethernet growing to make up ~20% of industry revenues
Scale up switching market revenue ($, mn)



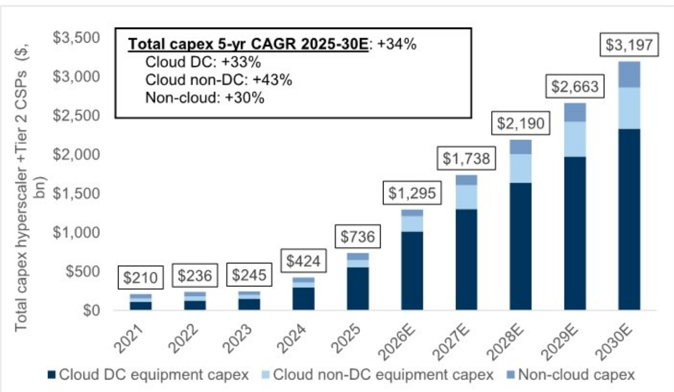
Source: 650 Group

Exhibit 12: 2026-2030 forecasts for cloud data center equipment capital expenditures were raised by ~28% on average relative to forecasts from 4Q25
Cloud DC equipment capital expenditures ($, mn)



Source: 650 Group, Data compiled by Goldman Sachs Global Investment Research

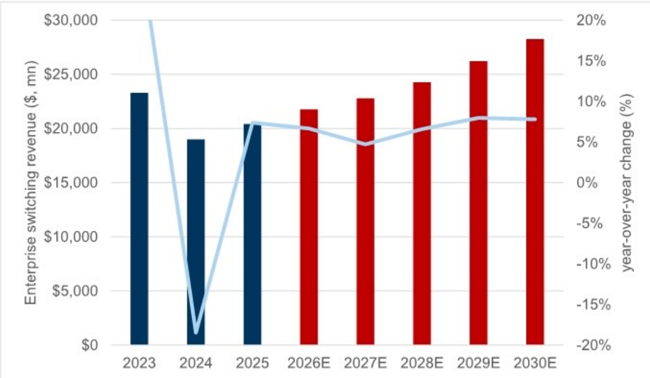
Exhibit 13: Cloud data center equipment capital expenditures are expected to grow at a 33% 2025-2030 CAGR to ~$3.2 trillion in 2030, in-line with total cloud provider capex growing at a 34% 5-year CAGR over the same period
Cloud capital expenditures over time ($, bn)



Source: 650 Group

Campus networking forecast updates

Exhibit 14: The enterprise switching market should grow 7% year-over-year in 2026, in-line its 7% 5-yr 2025-30 CAGR
Enterprise switching revenue ($, mn) & year-over-year change (%)

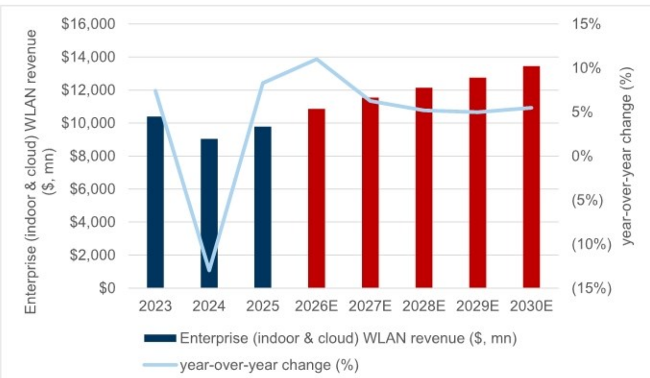


Source: 659 Group, Data compiled by Goldman Sachs Global Investment Research

Exhibit 15: Cisco continues to dominate the enterprise switching market
Enterprise switching market share by vendor (%)

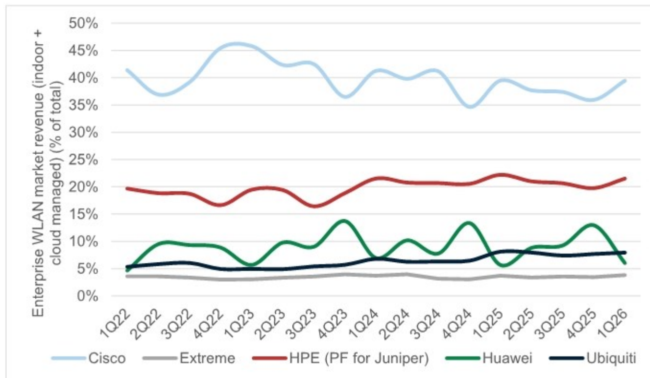
Source: 650 Group, Data compiled by Goldman Sachs Global Investment Research

Exhibit 16: The enterprise WLAN (indoor & cloud) market should grow 11% year-over-year in 2026, though should decelerate to MSD% annual growth thereafter
WLAN market revenue ($, mn) & year-over-year change (%)



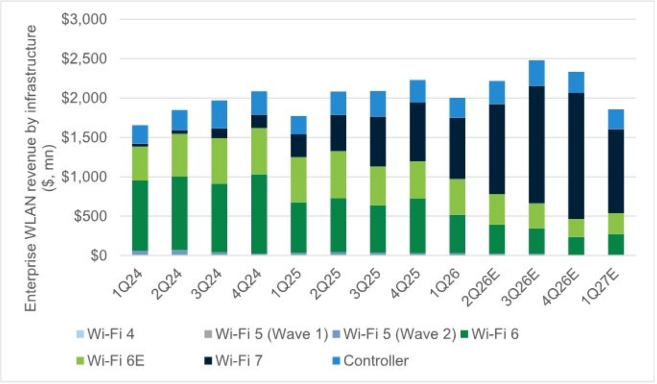
Source: 650 Group, Data compiled by Goldman Sachs Global Investment Research

Exhibit 17: Cisco leads the Enterprise WLAN (indoor & cloud) market at 36%, followed by HPE Aruba at 20%, Huawei (13%), and Ubiquiti (8%)
WLAN (indoor + cloud) market share by vendor (%)



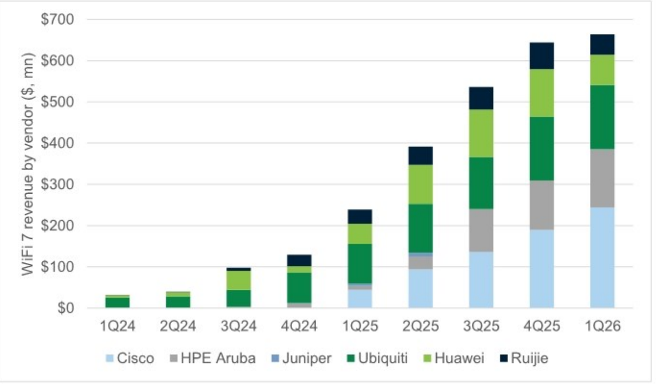
Source: 650 Group, Data compiled by Goldman Sachs Global Investment Research

Exhibit 18: Wi-Fi 7 grew its share of the total Enterprise WLAN market (including cloud) to 39% of total market revenue in 1Q26 and should grow its share towards ~60-70% by 4Q26-1Q27
Enterprise WLAN revenue by infrastructure type ($, mn)



Source: 650 Group, Data compiled by Goldman Sachs Global Investment Research

Exhibit 19: Within the WiFi 7 market, Cisco was the biggest sequential share gainer in 1Q26 (+6 pp to 31% share), followed by HPE which gained ~2 pp share (18%)
Wi-Fi 7 revenue by vendor ($, mn)

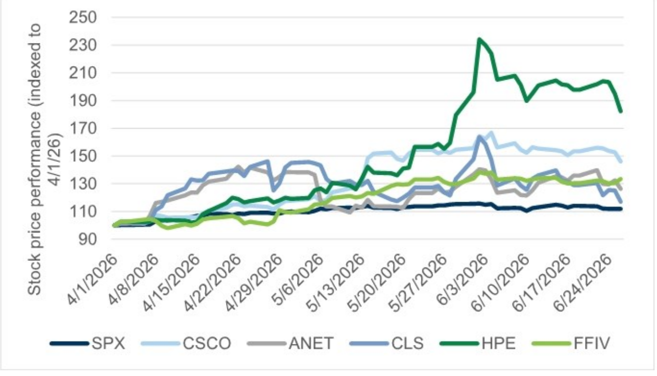


Includes select vendors

Source: 650 Group, Data compiled by Goldman Sachs Global Investment Research

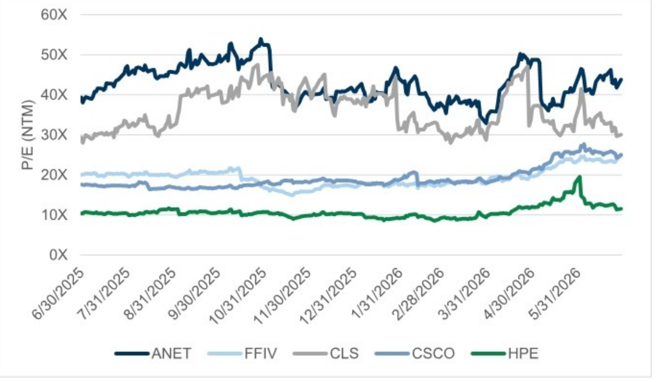
Historical stock performance around earnings

Exhibit 20: Across our networking coverage, all companies outperformed SPX
C2Q26 stock performance



Source: FactSet

Exhibit 21: With CLS most deviated from its 1Y average P/E (NTM) multiple
P/E (NTM) multiples across networking companies



Source: FactSet

Exhibit 22: Across our networking coverage, ANET has the strongest historical average performance in the two weeks leading up to earnings (but flat performance day of) while HPE has the strongest performance on the day following earnings (but worst average performance leading up to the print)
Historical stock performance around earnings

	Performance		
CSCO	**2 wks prior**	**Earnings**	**2 wks after**
F3Q26	(14.9%)	13.4%	(12.1%)
F2Q26	8.1%	(12.3%)	(7.7%)
F1Q26	(2.8%)	4.6%	(3.6%)
F4Q25	2.9%	(1.6%)	(3.0%)
Average	**(1.7%)**	**1.0%**	**(6.6%)**
ANET	**2 wks prior**	**Earnings**	**2 wks after**
C1Q26	20.2%	(13.6%)	1.6%
C4Q25	3.7%	4.8%	9.6%
C3Q25	24.4%	(8.6%)	(5.0%)
C2Q25	(11.0%)	17.5%	(7.1%)
Average	**9.3%**	**0.0%**	**(0.2%)**
CLS	**2 wks prior**	**Earnings**	**2 wks after**
C1Q26	10.8%	(14.4%)	(13.4%)
C4Q25	16.8%	(13.1%)	(10.2%)
C3Q25	(12.4%)	8.2%	(13.5%)
C2Q25	(15.3%)	16.5%	(6.4%)
Average	**(0.0%)**	**(0.7%)**	**(10.9%)**
HPE	**2 wks prior**	**Earnings**	**2 wks after**
F2Q26	(4.1%)	19.5%	(29.8%)
F1Q26	(2.3%)	(3.3%)	(8.3%)
F4Q25	(4.3%)	1.9%	(12.6%)
F3Q25	(7.7%)	1.5%	(7.8%)
Average	**(4.6%)**	**4.9%**	**(14.6%)**
FFIV	**2 wks prior**	**Earnings**	**2 wks after**
C1Q26	(0.3%)	3.4%	(4.0%)
C4Q25	(1.5%)	0.3%	(5.5%)
C3Q25	(6.1%)	2.0%	(7.2%)
C2Q25	(4.6%)	1.7%	2.2%
Average	**(3.1%)**	**1.8%**	**(3.6%)**

2 weeks prior & after refers to 10 workdays before and after the earnings print

Source: FactSet

Rating, price target, valuation, and key risks

Arista Networks Inc (ANET, Buy)

We are Buy-rated on ANET with a 12-month target price of $196 (unchanged) based on 36X (unchanged) our NTM+1Y EPS.

Key downside risks include slower cloud capex spending; customer concentration, with META and MSFT representing 16% and 26% of total revenue in 2025 respectively; competition, including from Cisco, major Chinese providers Huawei & H3C and whitebox switching solutions; margin degradation from investment in enterprise sales force, elevated costs associated with supply chain headwinds; broader pricing pressure from commoditization of networking hardware; inability to execute on enterprise/campus strategy & development of new SD-WAN platform.

Cisco Systems, Inc (CSCO, Neutral)

We are **Neutral-rated** on CSCO with a 12-month target price of $125 (unchanged) based on 25x (unchanged) our NTM+1Y (Q5-Q8) EPS.

Key upside risks include secular tailwinds including hybrid work, multi-cloud network architecture adoption, broader roll-out of WiFi 6/6E and 5G, increasing edge compute use cases; New disaggregated consumption models to target previously underserved cloud providers, near-term revenue visibility from elevated backlog; **downside risks** include competition, including from major Chinese providers Huawei & H3C and White Box solutions; margin degradation from mix shift into more cloud customers, elevated costs associated with supply chain headwinds; broader pricing pressure from commoditization of networking hardware; dilutive acquisitions.

Hewlett Packard Enterprise Co. (HPE, Buy)

We are **Buy rated** on HPE with a 12-month target price of $79 (unchanged) based on 18x (unchanged) our NTM+1Y EPS.

Key risks: (-) Lower-than-expected corporate IT spending and data center capex could result in worse-than-expected hardware sales; (-) Competition from white box manufacturers could result in cannibalization in HPE's server market share; (-) Market share loss from customer churn during Juniper integration; (-) Component costs could be higher than expected, resulting in worse gross margins; (-) Corporate actions from strategic investors; (-) Failure to deliver market share gains in storage driven by a strategy shift towards selling more owned-IP storage products that carry higher gross margins; (-) Worse-than-expected demand for enterprise and sovereign AI data center infrastructure.

Celestica Inc (CLS, Buy)

We are **Buy rated** on CLS with a 12-month target price of $475 (unchanged), reflecting 28X (unchanged) our NTM+1Y EPS.

Key downside risks include: (-) continued competition from branded data center equipment OEMs and ODMs, which could pressure market share; (-) AI server market competitive intensity, which can pressure AI server margins; (-) high customer concentration could cause revenue volatility; (-) potential declines in data center investment demand; (-) reductions in CLS's ability to win higher margin design-deals, which could limit CLS' ability to grow EBIAT margins over time; and (-) volatility in end-market demand within CLS' ATS segment.

Disclosure Appendix

Reg AC

I, Michael Ng, CFA, hereby certify that all of the views expressed in this report accurately reflect my personal views about the subject company or companies and its or their securities. I also certify that no part of my compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

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Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

Arista Networks Inc. (Buy, $169.88), Celestica Inc. (Buy, $364.80), Cisco Systems Inc. (Neutral, $117.46), F5 Inc. (Neutral, $415.96) and Hewlett Packard Enterprise Co. (Buy, $45.11).

The rating(s) for Arista Networks Inc., Celestica Inc., Cisco Systems Inc. and Hewlett Packard Enterprise Co. is/are relative to the other companies in its/their coverage universe: AT&T Inc., American Tower Corp., Apple Inc., Arista Networks Inc., Axon Enterprise Inc., Blend Labs, Celestica Inc., Charter Communications Inc., Cisco Systems Inc., Cogent Communications Holdings, Comcast Corp., Compass Inc., Crown Castle Inc., Dell Technologies Inc., Digital Realty Trust Inc., Equinix Inc., F5 Inc., HP Inc., Hewlett Packard Enterprise Co., IREN Ltd., Ingram Micro, Lumen Technologies Inc., NetApp Inc., Opendoor Technologies Inc., Optimum Communications Inc., Penguin Solutions Inc., SBA Communications Corp., Stagwell Inc., Super Micro Computer Inc., T-Mobile US Inc., TD SYNNEX Corp., Verizon Communications, Versant Media Group, Walt Disney Co., Zillow Group

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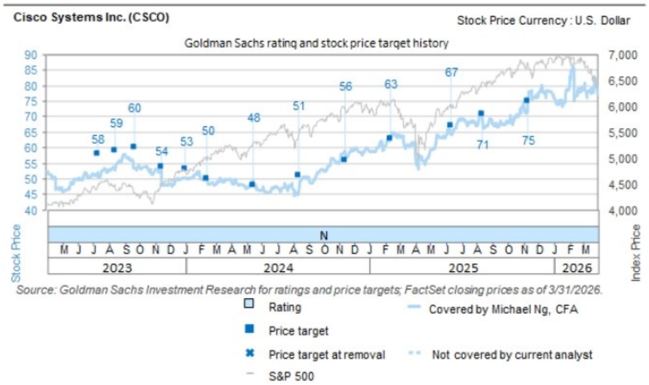
Price target and rating history chart(s)



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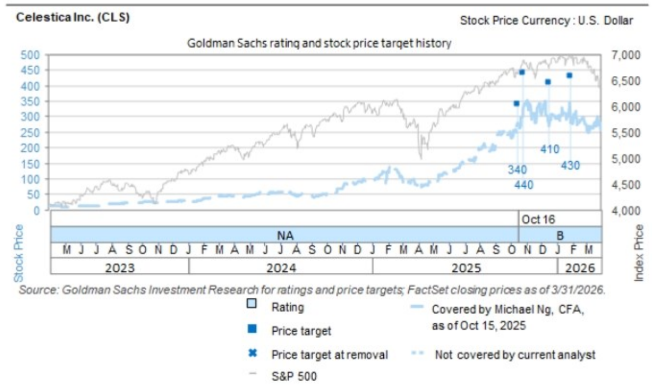

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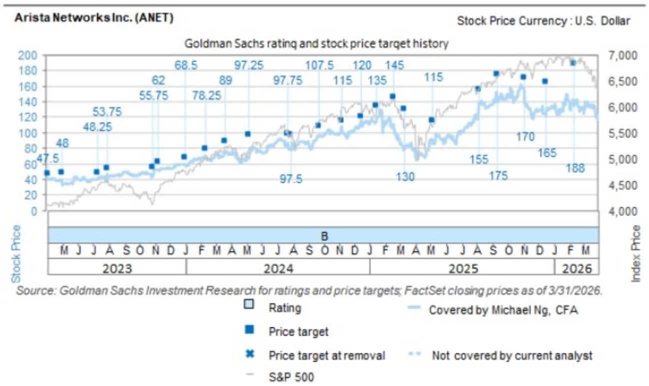

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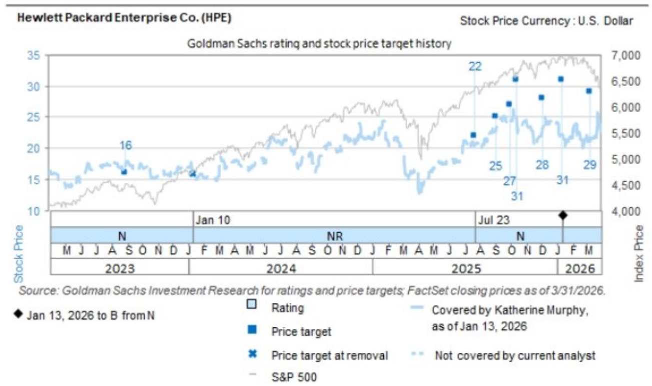

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Target price history table(s)

Hewlett Packard Enterprise Co. (HPE)

Date of report	Target price ($)	Closing price ($)
02-Jun-26	79.00	56.15
29-Apr-26	32.00	28.30
15-Apr-26	30.00	24.62
10-Mar-26	29.00	21.10
13-Jan-26	31.00	22.29
05-Dec-25	28.00	23.33

Cisco Systems Inc. (CSCO)

Date of report	Target price ($)	Closing price ($)
03-Jun-26	125.00	126.50
14-May-26	116.00	115.53
13-Nov-25	75.00	77.38
14-Aug-25	71.00	69.30
13-Jun-25	67.00	64.09
13-Feb-25	63.00	63.84

Hewlett Packard Enterprise Co. (HPE)

Date of report	Target price ($)	Closing price ($)
16-Oct-25	31.00	22.50
01-Oct-25	27.00	24.93
04-Sep-25	25.00	23.16
23-Jul-25	22.00	20.76
30-Aug-23	16.00	17.36

Cisco Systems Inc. (CSCO)

Date of report	Target price ($)	Closing price ($)
14-Nov-24	56.00	57.92
15-Aug-24	51.00	48.53
16-May-24	48.00	48.34
15-Feb-24	50.00	49.06
02-Jan-24	53.00	50.51
16-Nov-23	54.00	48.04
22-Sep-23	60.00	53.57
17-Aug-23	59.00	54.73
11-Jul-23	58.00	52.12

Arista Networks Inc. (ANET)

Date of report	Target price ($)	Closing price ($)
06-May-26	196.00	147.06
13-Feb-26	188.00	141.59
18-Dec-25	165.00	124.62
05-Nov-25	170.00	140.42
12-Sep-25	175.00	139.39
06-Aug-25	155.00	138.78
07-May-25	115.00	86.45
13-Mar-25	130.00	80.14
19-Feb-25	145.00	103.92
16-Jan-25	135.00	118.13
17-Dec-24	120.00	112.95
08-Nov-24	460.00	100.11
23-Sep-24	430.00	96.39
31-Jul-24	390.00	86.64
24-Jul-24	391.00	81.47
08-May-24	389.00	72.92
21-Mar-24	356.00	76.15
13-Feb-24	313.00	66.38
02-Jan-24	274.00	57.89
10-Nov-23	248.00	51.71
31-Oct-23	223.00	50.09
01-Aug-23	215.00	46.40
11-Jul-23	193.00	40.04

Celestica Inc. (CLS)

Date of report	Target price ($)	Closing price ($)
28-Apr-26	475.00	361.54
29-Jan-26	430.00	300.00
18-Dec-25	410.00	270.92
29-Oct-25	440.00	337.77
16-Oct-25	340.00	279.86

Price targets shown in table(s) are unadjusted for corporate actions.

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